



**C. V. Raman Global University
Odisha**

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Registration No. :

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Total number of printed pages – 03

**B.Tech
CS221**

**THIRD SEMESTER REGULAR EXAMINATION – 2024-25
DATA COMMUNICATION AND COMPUTER NETWORKS**

BRANCH: CSE AND ALLIED BRANCHES

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Illustrate the significance of signal-to-noise ratio (SNR) in data communication.	1	2	2
(b)	Which medium generally offers better security: guided or unguided? Justify your answer.	1	2	2
(c)	An analog signal has a frequency component of 15 kHz. If the signal is sampled at a rate of 20 kHz, compute the sampling frequency.	1	2	3
(d)	Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. Calculate the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference.	2	2	3
(e)	Mention the advantages of optical fiber over twisted pair and coaxial cable.	2	2	2
(f)	Calculate the bandwidth for AM if we need to modulate 5 KHz voice signal. Also state the total number of channels allocated by FCC for the same.	2	2	3
(g)	Draw the different HDLC frame formats and write each of their uses.	3	2	2
(h)	Briefly explain the types of network configurations provided by IEEE 802.11.	3	2	2
(i)	Find the minimum hamming distance for the given pair of words: (00111, 01011), (01011, 10101).	3	2	3

(j)	State the key differences between TCP and UDP.	4	2	2
(k)	Draw a labeled diagram of the structure of IPv4 datagram header.	4	2	2
(l)	Outline the functions of ARP and RARP in IPv4 network.	4	2	2
(m)	Name two mail access protocols and explain.	5	2	1
(n)	Discuss the relationship of HTTP with WWW.	5	2	2
(o)	Write two examples each for generic domain and country domain in DNS.	5	2	1

Part-B		CO	Marks	BL
Q2.	Answer any two:			
(a)	Describe each layer of the OSI model in detail with a neat diagram.	1	7	2
(b)	Encode the given data stream 11001110 using the following line coding schemes: (i) Unipolar NRZ (ii) Polar NRZ-I (iii) Differential Manchester (iv) AMI	1	7	3
(c)	(i) Explain how different network topologies impact data transmission efficiency. Compare between ring topology and mesh topology. (ii) How many direct connections (or links) are required for a fully connected mesh network with 5 nodes?	1	7	3
Q3.	Answer any two:			
(a)	i) Distinguish between synchronous and asynchronous time division multiplexing with relevant diagrams. ii) In a TDM system with 32 channels, each channel requires 8 bits/sample and the system operates at the rate of 8kHz. Calculate the total data rate required for this TDM system.	2	7	3
(b)	Discuss briefly the three different phases in a virtual circuit network with diagrams. State the role of virtual circuit identifier (VCI) in such a network.	2	7	2
(c)	Explain the need for modulation of digital data. Sketch the analog transmission form of the digital data sequence 1011011001 using BASK, BFSK and BPSK modulation techniques.	2	7	3
Q4.	Answer any two:			
a)	Differentiate between a random access protocol and a controlled access protocol. Briefly discuss the operation of CSMA/CD random access method with a suitable flow diagram.	3	7	2

(b)	Explain cyclic redundancy check (CRC). Given the data word 1010001101 and the divisor 110101, find the CRC and verify your answer.	3	7	3
(c)	Describe the working mechanism of Go-Back-N ARQ with suitable diagram. State its advantages over Stop-and-Wait ARQ.	3	7	2
Q5. Answer the followings:				
(a)	Define Congestion. Explain the principle and prevention policies of congestion control.			
(b)	For a given IP address 183.42.0.0/22, i) Identify the class of given IP address. ii) Write the default mask and customized subnet mask. iii) Find the number of hosts. iv) Find the number of networks. v) Find the valid subnets. vi) Find the broadcast address of each subnet.	4	7	2
Q6. Answer the followings:				
(a)	Discuss the client-server architecture. State the functionality of FTP with a suitable diagram.	5	7	2
(b)	List the different types of name servers. Explain the working of DNS briefly.	5	7	2

Best of Luck

3rd SEMESTER (SUMMER QUARTER) EXAMINATION – 2022
DATA COMMUNICATION & COMPUTER NETWORKS

BRANCH: ALL

Time: 3 Hours

Max Marks: 100

Answer all Questions from Part – A and Part – B
The figures in the right hand margin indicate marks.

PART – A

Q1. Answer the following questions:

[2x15]

- a) Illustrate protocol and its requirement in data communication.
- b) Describe the bit stuffing and its requirement in bit oriented protocol.
- c) Mention the four basic network topologies, and cite an advantage of each type.
- d) Discuss the carrier signal requirement for modulation. Write down the different digital to analog conversion techniques.
- e) Find the minimum hamming distance from the following code words.
i. d(10000, 00000) ii. d(10101, 10000) iii. d(00000, 11111)
- f) Define proxy server and explain its relation to HTTP.
- g) A channel has the bandwidth (B.W) of 50 MHz. Determine the channel capacity (C) for signal-to-noise ratio (SNR) of 70 dB.
- h) State the burst error? If the sent bit pattern is 1100010001000010 and received bit pattern is 1101110101100010, then find out the length of the burst error?
- i) A signal travels from point X to point Y. At point X, the signal power is 300 W. At point Y, the power is 250 W. Determine the attenuation in decibels.
- j) List out the basic differences between TCP and UDP.
- k) Explain the importance of ARP protocols in IP addressing.
- l) Illustrate the significance of the guard band in multiplexing? Five channels, each with a 50 kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 5 kHz between the channels to prevent interference?

- m) Define DNS. List out the different types of name servers.
- n) Mention the advantages of optical fiber over twisted pair and coaxial cable.
- o) State the two types of services provided by IEEE 802.11.

PART-B

Q2. (Answer any Two)

[7X2]

- a) Explain the functionalities of the seven layer OSI reference model with neat diagram. Illustrate the difference between OSI and TCP/IP model?
- b) Encode the given data word "0110101001" using NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester, Bipolar AMI and MLT-3 coding techniques.
- c) Explain the key features of FDM and TDM. Distinguish between synchronous and asynchronous TDM.

Q3. (Answer any Two)

[7X2]

- a) Explain the different stages of Pulse Code Modulation (PCM) technique with neat diagram to convert the analog signal to digital signal.
- b) Describe the operation of circuit switched network and packet switched network with suitable diagrams. Summarize the difference between circuit switched network and packet switched network.
- c) Discuss the importance of multiple accesses. Briefly explain the operations of CSMA/CD and CSMA/CA random access methods.

4. (Answer any Two)

[7X2]

- a) Explain the purpose of cyclic redundancy check (CRC). Given a 9-bit sequence 101100101 and a divisor polynomial x^4+x^2+1 , find the CRC code and verify your answer.
- b) Describe the mechanism of Stop and Wait ARQ. Distinguish between Go Back N ARQ and Selective Repeat ARQ.
- c) Find the following for the given IP address 194.32.47.40/24,
 - i. The network ID.
 - ii. The first host ID.
 - iii. The last host ID.
 - iv. The broad cast address.
 - v. The subnets.
 - vi. The valid hosts.
 - vii. The subnet mask address.

Q5. (Answer the following)

- a) Draw the schematic diagram of IPv4 header. Explain their various fields used in it. [7x2]
 Give the advantages of IPv6 over IPv4.
 b) Illustrate congestion control. Describe the principle and prevention policies of congestion control.

Q6. (Answer the following)

- a) Discuss the client-server model architecture? State the functionality of the FTP with a suitable diagram. [7x2]
 b) Explain the following:
 i. SMTP ii. WWW

Course Outcome Assessment Scheme:

CO1: Identify data communications system components, network topologies, and protocols.

CO2: Analyze different features of analog and digital transmission.

CO3: Analyze the working principles and protocols of data link layer.

CO4: Identify and differentiate working principles and protocols of network and transport layer.

CO5: Identify and implement different types of application in application layer.

COS	Questions	Total Marks
CO1	1(a), 1(c), 1(g), 1(i), 2(a), 2(b), 3(a)	29
CO2	1(d), 1(l), 1(n), 2(c), 3(b)	20
CO3	1(b), 1(e), 1(h), 1(o), 3(c), 4(a), 4(b)	29
CO4	1(j), 1(k), 4(c), 5(a), 5(b)	25
CO5	1(m), 1(f), 6(a), 6(b)	18

Question wise Cognitive Level distribution			
Cognitive Level	Questions		Percentage
Understand	1(a, b, c, d, j, m), 2(a, c), 3(a, b), 4(b), 6(b)	54	44.6 %
Apply	1(e, g, h, i, k, l, n, o), 2(b), 3(c), 4(a, c)	44	36.4%
Analyse	1(f), 5(a, b), 6(a)	23	19%

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B. Tech
CSE23305/CS20135

THIRD SEMESTER SUMMER QUARTER EXAMINATION – 2023
DATA COMMUNICATION AND COMPUTER NETWORKS
BRANCH: CSE/CSIT/CSSE/CSSYE/CC/DS/WT/IoTCS/CE/ AIML(LE)

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B
The figures in the right-hand margin indicate marks

PART – A

Q1. Answer the following Questions:

[2x15]

- Write different components of data communication system.
- Define the principal causes of transmission impairments.
- Mention the advantages of optical fiber over twisted pair cable.
- Discuss the principle of data communication followed in ring topology.
- Differentiate between thicknet and thin Ethernet.
- Draw a suitable diagram to display packet transmission in a packet switched network.
- Find the class, net-id and host-id of the following IP address: 155.25.45.48.
- A subnet has assigned a subnet mask of 255.255.255.192. Calculate maximum no. of host can be configured in the sub network.
- Differentiate between logical address and port address
- Compare and contrast between parallel and serial transmission mode.
- Find the minimum hamming distance between the following pair of code words:
(00000, 01011); (01011, 10101)
- Define flooding in routing Algorithm.
- List the different types of name servers used in DNS.
- Mention any four application layer protocols used in data communication.
- Illustrate the use of proxy server in HTTP.

PART-B

Q2. (Answer any two)

[7x2]

- Draw the OSI Model and compare the layer management of the OSI Reference Model with TCP/IP model.
- Show the NRZ-L, NRZ-I, RZ, Manchester and AMI encoding for the bit pattern given 01110011.
- Define topology. Explain different types of topologies with its relative advantages and disadvantages.

Q.3. (Answer any two)

[7x2]

- a. Define multiple access. Determine the maximum length of cable (in KM) for transmitting data rate 8000 MBPS in an Ethernet LAN with frames of size 5,000 bits. Assume the signal speed in the cable to be 4,00,000 m/s.
- b. Outline the mechanism of Stop and Wait ARQ. Distinguish between Go-Back-N ARQ and Selective Repeat ARQ.
- c. For transmitting data word 1011 find out the code word in hamming method if even parity method is used.

Q.4. (Answer any two)

[7x2]

- a. What do you mean by HDLC? Write down the services provided by HDLC and explain the frame structure in details.
- b. Explain Time division multiplexing technique with a neat diagram. Write the key differences between synchronous and statistical TDM.
- c. Compare and contrast a circuit switched network and packet switched networks.

Q.5. (Answer the followings)

[7x2]

- a. Differentiated between intra domain and inter domain routing. Explain the basic mechanism of distance vector routing with suitable example.
- b. Define Network Congestion. Explain different mechanisms of open-loop congestion control.

Q.6. (Answer the followings)

[7x2]

- a. Define domain name system (DNS). List out the different types of name servers. Explain the working of DNS briefly.
- b. Give short notes on HTTP and SMTP protocols

Course Outcome Assessment Scheme:

1. Identify data communications system components, network topologies, and protocols.
(Q1.a,b,c ; Q5.a,b)
2. Analyze different features of analog and digital transmission.
(Q1.d,e,f; Q3.a,b,c)
3. Analyze the working principles and protocols of data link layer.
(Q1.g,h,i,j ; Q2.a,b,c)
4. Identify and differentiate working principles and protocols of network and transport layer.
(Q1.k,l,m; Q4.a,b,c)
5. Identify and implement different types of application in application layer.
(Q1 n,o; Q6.a,b)



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B24

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**BTech
CSE23305**

THIRD SEMESTER BACKLOG EXAMINATION 2024-25
SUBJECT NAME: Data Communication and Computer Networks

BRANCH: CSIT

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Assume 8 devices are arranged in a mesh topology. Consider all devices using simplex mode for communication. Calculate the number of ports in each device and the number of cables required for the topology.	1	2	2
(b)	Describe Frequency Modulation (FM) with waveform.	1	2	2
(c)	What are the different transmission impairments?	1	2	2
(d)	A signal travels through a transmission medium and its power is reduced to one-half. Calculate the attenuation of the signal.	2	2	2
(e)	Define analog-to-analog conversion.	2	2	2
(f)	Define constellation diagram and explain its role in analog transmission.	2	2	1
(g)	Which of the three analog-to-analog conversion techniques (AM, FM, or PM) is the most susceptible to noise? Defend your answer.	3	2	2
(h)	Write Principles of Protocol Layering and explain logical connections with proper diagram.	3	2	2
(i)	Explain the concept of Encapsulation and Decapsulation with proper diagram.	3	2	1
(j)	Explain protocol. Give an example. Explain channelization.	4	2	2
(k)	Describe the need for framing.	4	2	3
(l)	Explain the difference between FDMA and CDMA.	4	2	3

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What is the Hamming distance for each of the following codewords?			
(m)	i. $d(10000, 00000)$ ii. $d(10101, 10000)$	5	2
(n)	A sender has two data items to send: $(4567)_{16}$ and $(BA98)_{16}$. What is the value of the checksum?	5	2
(o)	The maximum throughput of pure ALOHA is? Give derivation.	5	2

Q6.	Ans
(a)	Hov
(b)	Des

Part-B

		CO	Ma rks	Bl.
Q2. Answer any two:				
(a)	Encode the given data word "101100110111010" using NRZ-L, NRZ-I, Manchester, Differential Manchester, and AMI line coding techniques.	1	7	2
(b)	Show the different layers of OSI model with diagram. Write the key responsibility of each layer.	1	7	2
(c)	Distinguish data rate and signal rate. Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with four signal levels. Calculate the maximum bit rate.	1	7	2
Q3. Answer any two:				
(a)	Explain Internet Standards and maturity level of Request for Comment (RFC) using proper diagram.	2	7	2
(b)	Explain Binary Phase Shift Keying in detail.	2	7	2
(c)	Explain Block Coding and its use. A complex bandpass signal has a bandwidth of 200 kHz. What is the minimum sampling rate for this signal?	2	7	2
Q4. Answer any two:				
(a)	List out all the differences between Go-Back-N ARQ and Selective Repeat ARQ. Explain the both in detail.	3	7	2
(b)	Given the two, Flow and Error Control and Stop-and-wait ARQ, write down a short note for each.	3	7	2
(c)	Differentiate among various methods of multiple access.	3	7	2
Q5. Answer the followings:				
(a)	Explain in detail the following CSMA protocols. i. 1-persistent ii. p-persistent iii. Non-persistent	4	7	2
(b)	Write short notes on the following: i. Congestion control ii. Router	4	7	3